

JAD FALAQ

Machine Learning & MLOps Engineer – Deep Learning & Computer Vision

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6-month final-year engineering internship starting January 2027

PROFILE

Machine Learning Engineer, final-year AI engineering student at **ENSIAS**, with internships at **Renault Digital** and **AKWA Group**. From training to a **reliable model in production**, with an **ML pipeline on 6M+ transactions**, a **continuously deployed YOLOv8 detector** (DVC, MLflow, CI/CD, Azure) and **federated learning on 3D point clouds**. Specialized in **deep learning (PyTorch)**, **computer vision** and **MLOps**, focused on **rigorous evaluation** and **compact models**.

PROFESSIONAL EXPERIENCE

ML & Data Engineer – IT Data Team

AKWA Group | Internship | Casablanca, Morocco — Jun. 2025 – Sep. 2025

- **Missing transaction detection (CRISP-DM).** End-to-end ML pipeline on accounting entries from Afriware, the Odoo and JDE ERPs and internal files (6M+ transactions), with preprocessing, feature engineering and comparison of supervised (logistic regression, SVM, decision tree, Random Forest, KNN) and unsupervised models (Isolation Forest, autoencoder). Random Forest selected (98% accuracy, 85% F1-score) and integrated into the existing system.

AI Engineer Intern – Quality Engineering

Renault Digital | Internship | Casablanca, Morocco — Jun. 2026 – Sep. 2026

- **Multi-agent LLM system in production.** Assistant drafting Jira tickets from a need described in text, PDF or image, with parallel agents, RAG on past tickets and human validation before creation. Copilot Studio prototype then full stack (FastAPI, React/TypeScript), writing time cut from 10–20 min to 1–2 min per ticket.
- **Evaluation & reliability.** Calibrated similarity threshold with fallback on model failure, outputs validated by Pydantic and the Cucumber parser, sensitive-data masking, 85% of generations accepted without edits, in Agile sprints.

TECHNICAL SKILLS

- **Deep learning.** PyTorch, TensorFlow/Keras, CNN, U-Net, LSTM with attention, encoder-decoder, Transformers (BERT fine-tuning), GNN (PyTorch Geometric), transfer learning, knowledge distillation, federated learning (FedAvg)
- **Computer vision.** YOLOv8, Faster R-CNN, OpenCV, torchvision, DeepSORT, MediaPipe Pose, semantic segmentation, real-time
- **Machine learning.** Logistic regression, SVM, decision trees, Random Forest, XGBoost, PCA, LDA, t-SNE, autoencoders
- **Training & evaluation.** Cross-validation, grid search, early stopping, weighted loss, class imbalance, data augmentation, threshold calibration, external splits, precision, recall, F1, mAP, mIoU, ROC-AUC, confusion matrix, error analysis
- **Data & signal.** Python, SQL, Pandas, NumPy, Seaborn, feature extraction, up/downsampling, data leakage, time series
- **MLOps & deployment.** MLflow (tracking, model registry), DVC, Docker, GitHub Actions (CI/CD), pytest, FastAPI, Azure
- **Inference optimization.** CUDA/GPU, real-time inference, compact models (distillation), optimized Docker image, Accelerate

EDUCATION

State Engineering Degree – Artificial Intelligence Engineering (2IA)

ENSIAS, Rabat | Sep. 2024 – Jun. 2027

- Machine learning and deep learning (neural networks, reinforcement learning), statistical and probabilistic learning
- Big Data and distributed systems, data engineering, time series, multimedia processing (text, image)

Preparatory Classes for the French Grandes Écoles (CPGE) – Mathematics & Physics

CPGE Mohammed V, Casablanca | Sep. 2022 – Jun. 2024

ACADEMIC PROJECTS

- **Dental pathology detection – End-to-end MLOps.** YOLOv8 on 1,808 X-rays (24,891 audited annotations, 14 classes remapped into 7), 3 variants compared on validation, test and external sets in MLflow, model selected for its recall then continuously deployed (DVC, FastAPI, Docker image reduced from 3.74 to 1.19 GB, CI/CD, Azure)
- **AmnAI – Real-time theft detection.** YOLO detection, DeepSORT tracking and MediaPipe pose, then behavioral features classified by a bidirectional LSTM with attention, 80–92% precision at 20–28 FPS versus 65–75% for the heuristic baseline
- **PointFed3D – Federated learning on 3D point clouds.** 5 training strategies compared for a FeaStNet GNN on ShapeNet (50 classes), centralized (82.5%), FedAvg on 4 IID (79.8%) and non-IID clients, vertical federation and a 7.7× smaller distilled student
- **Medical NLP – AraBERT fine-tuning.** Medical misinformation detection in Arabic tweets with weighted loss, early stopping and threshold tuning, 87% accuracy and 0.87 macro F1 against a TF-IDF baseline

SOFT SKILLS

- **Entrepreneurial mindset.** Automations launched on my own initiative, e-commerce websites built for clients
- **Listening & ownership.** Product shaped by user feedback, no AI action without human validation

LANGUAGES

French (native) – Arabic (native) – English (bilingual)